CCNA 200-301 Notes

Chapter 05 - IPv4 Addressing Fundamentals

Learning Objectives

Understand IPv4, Static vs DHCP, IPv4 Classes, Loopback,
Network ID vs Host ID, and common CCNA exam mistakes.

1. IPv4 Basics

• Every device needs a unique IPv4 Address.
• IPv4 = 32 Bits.
• 4 Octets.
• 8 Bits per Octet.
• Written in Decimal, stored internally in Binary.
Example: 192.168.1.10

2. Static IP vs DHCP

Static IP:
- Configured manually.
- Common for servers, routers and printers.

DHCP:
- Assigns IP automatically.
- Also provides Subnet Mask, Default Gateway and DNS.

3. IANA & IPv4 Classes

IANA organized IPv4 addresses into classes.

Class A : 1-126 Mask 255.0.0.0 ~16.7 Million Hosts
Class B : 128-191 Mask 255.255.0.0 ~65,534 Hosts
Class C : 192-223 Mask 255.255.255.0 254 Hosts

127.x.x.x = Loopback
224-239 = Multicast
240-255 = Experimental

4. Loopback Address

127.0.0.1 is used to test the local TCP/IP stack.
Command:
ping 127.0.0.1

It does NOT test the cable, switch, router or Internet.

5. Network ID vs Host ID

Example:
IP: 192.168.1.10
Mask: 255.255.255.0

Network ID = 192.168.1
Host ID = 10

Rule:
255 -> Network
0 -> Host

6. Number of Hosts

Class A -> $2^{24} - 2 \approx$ 16.7 Million
Class B -> $2^{16} - 2$ = 65,534
Class C -> $2^{8} - 2$ = 254

Why -2?
One address is reserved for the Network Address and one for the Broadcast Address.

7. Common Exam Mistake

Never determine the Class from the Subnet Mask.
Always determine it from the FIRST OCTET.

Example:
11.15.200.7 / 255.255.255.0
Answer: Class A.

Exam Tips

✓ Memorize the class ranges.
✓ Memorize the default masks.
✓ 127.0.0.1 = Loopback.
✓ First Octet determines the Class.
✓ DHCP = Automatic, Static = Manual.

30-Second Revision

IPv4 = 32 Bits
4 Octets
8 Bits / Octet
Class A = 1-126
Class B = 128-191

Class C = 192-223
127.0.0.1 = Loopback
255 = Network
0 = Host
Class = First Octet